

Tanu Chaudhary

+91 6397113386 | [Email](#) | [LinkedIn](#) | [GitHub](#)

SUMMARY

Software Engineer with strong foundations in Data Structures, Algorithms, OOPs, DBMS, and system design, experienced in developing scalable backend systems, distributed applications, and REST APIs using Java, Python, Spring Boot, FastAPI, and MERN technologies, with a strong interest in problem-solving and software development.

EDUCATION

Vellore Institute of Technology, Chennai

Master of Computer Applications CGPA: 8.62

Chennai, Tamil Nadu

Expected Graduation 2027

Galgotias University, Noida, UP

Bachelor of Computer Applications (AIML) CGPA: 8.6

2022-2025

INTERNSHIP & TRAINING

Google Generative AI Virtual Intern | AICTE

07/2024 – 09/2024

- Applied prompt engineering for real-time intelligent responses and automation
- Developed foundational skills in generative AI and AI automation concepts.

RESEARCH & ACHIEVEMENTS

- Research: Quantum-Inspired Multi-Agent Learning for Blue Carbon Modeling (Under Review – Springer)
- Smart India Hackathon: Pre-Qualifier (2023, 2024)

PROJECTS

Real-Time Disaster Monitoring & Emergency Response System | [Github](#)

Tech Stack: Java, Spring Boot, MySQL, REST APIs

- Engineered scalable backend services for real-time emergency alert processing and incident management using Spring Boot.
- Designed optimized REST API workflows and database handling for reliable high-volume request processing.

Smart City Management & Complaint Resolution Platform | [Github](#)

Tech Stack: MERN Stack (MongoDB, Express.js, React.js, Node.js)

- Built a scalable MERN-based platform for real-time complaint tracking and civic service management.
- Developed RESTful APIs and optimized backend workflows for efficient concurrent request handling and system scalability

Crime Prediction System | [Github](#)

Tech Stack: Python, FastAPI, XGBoost, MongoDB

- Built FastAPI-based backend pipelines for crime data ingestion, preprocessing, and real-time predictive analysis.
- Integrated machine learning models with scalable backend services, improving prediction processing efficiency by 25%.

SKILLS

Programming Languages: Java, Python, C++, JavaScript, C, PHP

Frontend Development: React.js, Angular, HTML, CSS, AJAX, jQuery

Backend & Systems: Node.js, Spring Boot, FastAPI, Express.js, REST APIs, Distributed Systems, Microservices, Concurrency, Multi-threading, System Design, MySQL, MongoDB

Core Computer Science: Data Structures & Algorithms, OOPs, DBMS, Operating Systems, Computer Networks

Machine Learning & AI: Machine Learning, Deep Learning, Data Preprocessing, Model Evaluation, Transformers

Tools & Platforms: Git, GitHub, Docker, AWS (EC2, S3), OpenStack, Postman, VS Code

CERTIFICATES

Web Development Fundamentals – IBM [Link](#)

Database Programming with SQL – Oracle [Link](#)

Microsoft AI & ML Engineering – Microsoft [Link](#)

AWS Machine Learning Foundations – AWS [Link](#)